

Homework 02

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Problem 1. Let X , Y and Z be random variables and let c be a number. Prove that

$$\text{Var}(cX) = c^2 \text{Var}(X),$$

$$\text{Cov}(cX, Y) = c \text{Cov}(X, Y)$$

and

$$\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z).$$

(i) $\text{Var}(cX) = c^2 \text{Var}(X)$.

$$\begin{aligned} \text{Var}(cX) &= \mathbb{E}\left((cX - \mathbb{E}(cX))^2\right) \\ &= \mathbb{E}\left(c^2 X^2 - 2cX \mathbb{E}(cX) + (\mathbb{E}(cX))^2\right) \\ &= \mathbb{E}\left(c^2 X^2 - 2c^2 X \mathbb{E}(X) + (c \mathbb{E}(X))^2\right) \\ &= \mathbb{E}\left(c^2 X^2 - 2c^2 X \mathbb{E}(X) + c^2 (\mathbb{E}(X))^2\right) \\ &= \mathbb{E}\left(c^2 \left(X^2 - 2X \mathbb{E}(X) + (\mathbb{E}(X))^2\right)\right) \\ &= c^2 \mathbb{E}\left(X^2 - 2X \mathbb{E}(X) + (\mathbb{E}(X))^2\right) \\ &= c^2 \mathbb{E}\left((X - \mathbb{E}(X))^2\right) \\ &= c^2 \text{Var}(X). \end{aligned}$$

(ii) $\text{Cov}(cX, Y) = c \text{Cov}(X, Y)$. Starting from the deviation definition $\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$, and using $\mu_{cX} = c \mu_X$:

$$\begin{aligned} \text{Cov}(cX, Y) &= \mathbb{E}[(cX - \mu_{cX})(Y - \mu_Y)] \\ &= \mathbb{E}[(cX - c\mu_X)(Y - \mu_Y)] \\ &= \mathbb{E}[c(X - \mu_X)(Y - \mu_Y)] \\ &= c \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \\ &= c \text{Cov}(X, Y). \end{aligned}$$

(iii) $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$.

$$\begin{aligned}
 \text{Cov}(X + Y, Z) &= \mathbb{E}[(X + Y - \mu_{X+Y})(Z - \mu_Z)] \\
 &= \mathbb{E}[(X + Y - \mu_X - \mu_Y)(Z - \mu_Z)] \\
 &= \mathbb{E}[(X - \mu_X) + (Y - \mu_Y)](Z - \mu_Z)] \\
 &= \mathbb{E}[(X - \mu_X)(Z - \mu_Z) + (Y - \mu_Y)(Z - \mu_Z)] \\
 &= \mathbb{E}[(X - \mu_X)(Z - \mu_Z)] + \mathbb{E}[(Y - \mu_Y)(Z - \mu_Z)] \\
 &= \text{Cov}(X, Z) + \text{Cov}(Y, Z).
 \end{aligned}$$

The collapse $\mu_{X+Y} = \mu_X + \mu_Y$ on the second line, and the split of the expectation on the fifth, are both linearity of expectation — which is exactly what Problem 2 proves.

Problem 2. Let X_1 and X_2 be discrete random variables and suppose that a_1 and a_2 are real numbers. Using the joint probability mass function for X_1 and X_2 , show that

$$\mathbb{E}[a_1X_1 + a_2X_2] = a_1\mathbb{E}[X_1] + a_2\mathbb{E}[X_2].$$

Write $p(x_1, x_2)$ for the joint probability mass function of X_1 and X_2 , and p_{X_1} , p_{X_2} for the marginals.

$$\begin{aligned}
 \mathbb{E}[a_1X_1 + a_2X_2] &= \sum_{x_1} \sum_{x_2} (a_1x_1 + a_2x_2) p(x_1, x_2) \\
 &= \sum_{x_1} \sum_{x_2} a_1x_1 p(x_1, x_2) + \sum_{x_1} \sum_{x_2} a_2x_2 p(x_1, x_2) \\
 &= \sum_{x_1} a_1x_1 p_{X_1}(x_1) + \sum_{x_2} \sum_{x_1} a_2x_2 p(x_1, x_2) \\
 &= \sum_{x_1} a_1x_1 p_{X_1}(x_1) + \sum_{x_2} a_2x_2 p_{X_2}(x_2) \\
 &= a_1 \sum_{x_1} x_1 p_{X_1}(x_1) + a_2 \sum_{x_2} x_2 p_{X_2}(x_2) \\
 &= a_1\mathbb{E}[X_1] + a_2\mathbb{E}[X_2].
 \end{aligned}$$

The third and fourth lines are marginalization: summing the joint pmf over the variable that does not appear in the summand leaves the marginal pmf of the other.

Problem 9. (Edition 11, Chapter 2.) If the distribution function of X is given by

$$F(b) = \begin{cases} 0 & b < 0, \\ \frac{1}{2} & 0 \leq b < 1, \\ \frac{3}{5} & 1 \leq b < 2, \\ \frac{4}{5} & 2 \leq b < 3, \\ \frac{9}{10} & 3 \leq b < 3.5, \\ 1 & b \geq 3.5, \end{cases}$$

calculate the probability mass function of X .

The mass at each point is the size of the jump of F there, so subtract consecutive values of F : $p(1) = F(1) - F(0)$, and so on.

$$p(0) = \frac{1}{2}, \quad p(1) = \frac{1}{10}, \quad p(2) = \frac{1}{5}, \quad p(3) = \frac{1}{10}, \quad p(3.5) = \frac{1}{10},$$

and $p(x) = 0$ for every other x .

Problem 12. (Edition 11, Chapter 2.) On a multiple-choice exam with three possible answers for each of the five questions, what is the probability that a student would get four or more correct answers just by guessing?

Five questions, three options each, so the number correct is Binomial(5, 1/3) and $\mathbb{P}(\text{correct}) = \frac{1}{3}$ on any one question. Four or more correct means four or five:

$$\begin{aligned}\mathbb{P}(4) + \mathbb{P}(5) &= \binom{5}{4} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right) + \left(\frac{1}{3}\right)^5 \\ &= \frac{10}{243} + \frac{1}{243} = \boxed{\frac{11}{243}}.\end{aligned}$$

Problem 22. (Edition 11, Chapter 2.) If a fair coin is successively flipped, find the probability that a head first appears on the fifth trial.

A first head on the fifth trial is exactly the one sequence $TTTTH$, and the flips are independent and fair, so

$$\mathbb{P}(\text{first head on trial 5}) = \left(\frac{1}{2}\right)^5 = \frac{1}{32}.$$

Problem 37. (Edition 11, Chapter 2.) Let $X_1, X_2, \dots, X_n$ be independent random variables, each having a uniform distribution over $(0, 1)$. Let $M = \text{maximum}(X_1, X_2, \dots, X_n)$. Show that the distribution function of M , $F_M(\cdot)$, is given by

$$F_M(x) = x^n, \quad 0 \leq x \leq 1.$$

What is the probability density function of M ?

The maximum is at most x exactly when *every* X_i is at most x . Concretely, for $n = 3$ and $x = \frac{1}{2}$, the event is $X_1 \leq \frac{1}{2}, X_2 \leq \frac{1}{2}, X_3 \leq \frac{1}{2}$, which by independence has probability $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \left(\frac{1}{2}\right)^3$. In general, using independence and $\mathbb{P}(X_i \leq x) = x$ for $0 \leq x \leq 1$,

$$F_M(x) = \mathbb{P}(M \leq x) = \mathbb{P}(X_1 \leq x) \mathbb{P}(X_2 \leq x) \cdots \mathbb{P}(X_n \leq x) = x^n, \quad 0 \leq x \leq 1.$$

The curve runs from 0 to 1 and is steeper near 1 as n grows. Differentiating,

$$f_M(x) = \frac{d}{dx} [x^n] = nx^{n-1}, \quad 0 \leq x \leq 1.$$

Problem 61. (Edition 11, Chapter 2.) Let $X_1, X_2, \dots$ be a sequence of independent, identically distributed continuous random variables. We say that a record occurs at time n if $X_n > \max(X_1, \dots, X_{n-1})$ — that is, if X_n is larger than each of $X_1, \dots, X_{n-1}$. Show that

1. $P\{\text{a record occurs at time } n\} = 1/n$;
2. $\mathbb{E}[\text{number of records by time } n] = \sum_{i=1}^n 1/i$;
3. $\text{Var}(\text{number of records by time } n) = \sum_{i=1}^n (i-1)/i^2$;
4. letting $N = \min\{n : n > 1 \text{ and a record occurs at time } n\}$, $\mathbb{E}[N] = \infty$.

Hint for (b) and (c): represent the number of records as a sum of indicator variables.

(a) $P\{\text{a record occurs at time } n\} = 1/n$. The X_i are continuous, so $P\{X_i = X_j\} = 0$ for $i \neq j$ and with probability 1 the n values are distinct and can be ranked. Because $X_1, \dots, X_n$ are independent and identically distributed, relabelling them leaves the joint distribution unchanged, so each of the $n!$ orderings of $X_1, \dots, X_n$ from smallest to largest carries the same probability $1/n!$.

A record at time n says exactly one thing about the ordering: X_n is larger than every other value, so it sits last. Fixing X_n in the last place leaves the remaining $n - 1$ values to arrange freely, which happens in $(n - 1)!$ ways. Hence

$$P\{X_n > \max(X_1, \dots, X_{n-1})\} = \frac{(n-1)!}{n!} = \frac{1}{n}.$$

Check at $n = 2$, by integration. With common density f and distribution function F ,

$$P\{X_2 > X_1\} = \int_{-\infty}^{\infty} \int_{-\infty}^{x_2} f(x_1) f(x_2) dx_1 dx_2 = \int_{-\infty}^{\infty} f(x_2) F(x_2) dx_2.$$

Integrating by parts with $u = F(x_2)$ and $dv = f(x_2) dx_2$,

$$\int_{-\infty}^{\infty} f(x_2) F(x_2) dx_2 = \left[F(x_2)^2 \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f(x_2) F(x_2) dx_2,$$

so twice the integral equals $F(\infty)^2 - F(-\infty)^2 = 1 - 0 = 1$, and

$$\int_{-\infty}^{\infty} f(x_2) F(x_2) dx_2 = \left[\frac{F(x_2)^2}{2} \right]_{-\infty}^{\infty} = \frac{1}{2},$$

which is $1/n$ at $n = 2$. The counting argument above proves every n directly, so no induction is needed.

(b) $\mathbb{E}[\text{number of records by time } n] = \sum_{i=1}^n 1/i$. Let

$$I_i = \begin{cases} 1 & \text{if a record occurs at time } i, \\ 0 & \text{otherwise,} \end{cases}$$

so that I_i is a random variable taking the values 0 and 1, and the number of records by time n is the sum $N = \sum_{i=1}^n I_i$. Each record contributes exactly 1 to that sum and each non-record contributes 0, so the sum counts them.

$$\begin{aligned} \mathbb{E}[\text{number of records by time } n] &= \mathbb{E}\left[\sum_{i=1}^n I_i\right] \\ &= \sum_{i=1}^n \mathbb{E}[I_i] \\ &= \sum_{i=1}^n \left(\frac{1}{i} \cdot 1 + \frac{i-1}{i} \cdot 0 \right) \\ &= \sum_{i=1}^n \frac{1}{i}. \end{aligned}$$

The second line is linearity of expectation, proved in Problem 2. The third evaluates $\mathbb{E}[I_i]$ directly from the definition of expectation: I_i takes the value 1 with probability $1/i$ by part (a), and the value 0 with the remaining probability $(i - 1)/i$.

(c) $\text{Var}(\text{number of records by time } n) = \sum_{i=1}^n (i-1)/i^2$. Keep the indicators I_i and the identity $N = \sum_{i=1}^n I_i$ from part (b).

$$\begin{aligned}
 \text{Var}(N) &= \text{Var}\left(\sum_{i=1}^n I_i\right) \\
 &= \sum_{i=1}^n \text{Var}(I_i) \\
 &= \sum_{i=1}^n \left(\mathbb{E}[I_i^2] - \mathbb{E}[I_i]^2\right) \\
 &= \sum_{i=1}^n \left(1^2 \cdot \frac{1}{i} + 0^2 \cdot \frac{i-1}{i} - \left(1 \cdot \frac{1}{i} + 0 \cdot \frac{i-1}{i}\right)^2\right) \\
 &= \sum_{i=1}^n \left(\frac{1}{i} - \frac{1}{i^2}\right) = \sum_{i=1}^n \frac{i-1}{i^2}.
 \end{aligned}$$

The third line is the definition of variance. The fourth evaluates both moments from the distribution of I_i , which takes the value 1 with probability $1/i$ by part (a) and the value 0 with probability $(i-1)/i$; squaring a 0/1 variable leaves it unchanged, so $\mathbb{E}[I_i^2] = \mathbb{E}[I_i] = 1/i$.

The second line is additivity of variance over a sum. It is not linearity, and it is not free: in general $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X,Y)$, so dropping the cross terms requires $\text{Cov}(I_i, I_j) = 0$ for every $i \neq j$. The indicators are in fact independent, which gives that. Fix k and condition on the relative ranking of $X_1, \dots, X_{k-1}$, which is what determines $I_1, \dots, I_{k-1}$. All $k!$ orderings of $X_1, \dots, X_k$ are equally likely, and each ranking of the first $k-1$ values is refined by exactly k of them, one for each slot X_k can occupy among the k . So X_k is the largest of the k with probability $1/k$ whatever the earlier ranking was:

$$P\{I_k = 1 \mid I_1 = \varepsilon_1, \dots, I_{k-1} = \varepsilon_{k-1}\} = \frac{1}{k} = P\{I_k = 1\}$$

for every choice of $\varepsilon_1, \dots, \varepsilon_{k-1}$ in $\{0, 1\}$ that can occur. Applying this at $k = n, n-1, \dots, 2$ factors the joint distribution, so $I_1, \dots, I_n$ are independent and every covariance vanishes. For $n = 3$ this is visible directly in the six orderings: $P\{I_2 = 1\} = 1/2$, $P\{I_3 = 1\} = 1/3$, and $P\{I_2 = 1, I_3 = 1\} = 1/6$, which is the product.

(d) $N = \min\{n : n > 1 \text{ and a record occurs at time } n\}$ has $\mathbb{E}[N] = \infty$. N takes values in $\{2, 3, \dots\}$, since time 1 is a record by convention and N is the first record time after it. So

$$\mathbb{E}[N] = \sum_{i=2}^{\infty} i P\{N = i\}.$$

Fix $i \geq 2$ and count orderings of $X_1, \dots, X_i$, of which all $i!$ are equally likely. The event $\{N = i\}$ says two things: no record occurred at any of the times $2, \dots, i-1$, which means nothing among $X_2, \dots, X_{i-1}$ ever overtook X_1 , so X_1 is the largest of $X_1, \dots, X_{i-1}$; and a record occurred at time i , so X_i is the largest of all i . Those two constraints pin the last two places in the ordering — X_i largest, X_1 next — and leave the remaining $i-2$ values to be arranged freely, in $(i-2)!$ ways. Hence

$$P\{N = i\} = \frac{(i-2)!}{i!}.$$

Substituting, and using $i! = i(i-1)!$ to cancel the leading i ,

$$\mathbb{E}[N] = \sum_{i=2}^{\infty} i \frac{(i-2)!}{i!} = \sum_{i=2}^{\infty} \frac{(i-2)!}{(i-1)!} = \sum_{i=2}^{\infty} \frac{1}{i-1} = \sum_{m=1}^{\infty} \frac{1}{m},$$

the harmonic series, which diverges. So $\mathbb{E}[N] = \infty$.

Problem 76. (*Edition 11, Chapter 2.*) Use Chebyshev's inequality to prove the weak law of large numbers. Namely, if $X_1, X_2, \dots$ are independent and identically distributed with mean μ and variance σ^2 , then for any $\varepsilon > 0$,

$$P\left\{\left|\frac{X_1 + X_2 + \dots + X_n}{n} - \mu\right| > \varepsilon\right\} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Write $\bar{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n}$ for the sample mean. Chebyshev's inequality says that for a random variable Y with finite mean and variance and any $k > 0$,

$$\mathbb{P}(|Y - \mathbb{E}[Y]| \geq k) \leq \frac{\text{Var}(Y)}{k^2},$$

so the two things needed before it can be applied to $\bar{X}_n$ are $\mathbb{E}[\bar{X}_n]$ and $\text{Var}(\bar{X}_n)$.

The mean is μ . By linearity of expectation,

$$\mathbb{E}[\bar{X}_n] = \frac{1}{n} \mathbb{E}[X_1 + \dots + X_n] = \frac{1}{n} (\mathbb{E}[X_1] + \dots + \mathbb{E}[X_n]) = \frac{n\mu}{n} = \mu,$$

so the quantity inside the absolute value in the statement is exactly $\bar{X}_n - \mathbb{E}[\bar{X}_n]$, which is the form Chebyshev requires.

The variance shrinks like $1/n$. The constant $1/n$ leaves a variance squared, and because the X_i are independent the variance of the sum is the sum of the variances:

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Now fix $\varepsilon > 0$ and apply Chebyshev to $Y = \bar{X}_n$ with $k = \varepsilon$. The event in the problem is the strict one, $|\bar{X}_n - \mu| > \varepsilon$, and it is contained in the non-strict event $|\bar{X}_n - \mu| \geq \varepsilon$ that Chebyshev bounds, so the bound applies to it as well:

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}.$$

With σ^2 and ε both fixed, the right-hand side is a constant divided by n , so it tends to 0 as $n \rightarrow \infty$ whatever the value of ε . A probability is non-negative, so the left-hand side is squeezed to 0 as well:

$$\mathbb{P}\left(\left|\frac{X_1 + X_2 + \dots + X_n}{n} - \mu\right| > \varepsilon\right) \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

which is the weak law of large numbers.

Problem 86. (*From Edition 11.*) Each new book donated to a library must be processed. Suppose that the time it takes a librarian to process a book has mean 10 minutes and standard deviation 3 minutes. If a librarian has 40 books that must be processed one at a time,

1. approximate the probability that it will take more than 420 minutes to process all these books;
2. approximate the probability that at least 25 books will be processed in the first 240 minutes.

Let T_i be the time taken to process book i . The T_i are independent and identically distributed with $\mathbb{E}[T_i] = 10$ minutes and $\text{SD}(T_i) = 3$ minutes, so $\text{Var}(T_i) = 9 \text{ min}^2$. Because the books are processed one at a time, the moment book n is finished is

$$S_n = T_1 + T_2 + \cdots + T_n,$$

and by linearity of expectation and independence,

$$\mathbb{E}[S_n] = 10n, \quad \text{Var}(S_n) = 9n, \quad \text{SD}(S_n) = 3\sqrt{n}.$$

The standard deviation is the square root of the variance of the sum; it is not n times $\text{SD}(T_i)$. By the central limit theorem, for n this large

$$Z = \frac{S_n - \mathbb{E}[S_n]}{\text{SD}(S_n)}$$

is approximately standard normal, and $\Phi(z) = \mathbb{P}(Z \leq z)$ denotes the standard normal distribution function.

(a) More than 420 minutes to process all 40 books. All 40 books are finished at time S_{40} , with

$$\begin{aligned} \mathbb{E}[S_{40}] &= 10(40) = 400 \text{ minutes}, & \text{Var}(S_{40}) &= 9(40) = 360 \text{ min}^2, \\ \text{SD}(S_{40}) &= \sqrt{360} = 6\sqrt{10} \approx 18.97. \end{aligned}$$

Standardizing 420,

$$z = \frac{420 - 400}{6\sqrt{10}} = \frac{20}{6\sqrt{10}} = \frac{10}{3\sqrt{10}} = \frac{\sqrt{10}}{3} \approx 1.054.$$

The event is “more than,” so it is the upper tail:

$$\begin{aligned} \mathbb{P}(S_{40} > 420) &\approx \mathbb{P}\left(Z > \frac{\sqrt{10}}{3}\right) = 1 - \Phi\left(\frac{\sqrt{10}}{3}\right) \\ &= 1 - 0.8541 = \boxed{0.1459}. \end{aligned}$$

(b) At least 25 books processed in the first 240 minutes. The normal approximation applies to a sum of processing times, not to a count of books, so the event must first be rewritten as a statement about S_{25} . At least 25 books are finished by minute 240 precisely when the 25th book is finished by minute 240, that is when $S_{25} \leq 240$. Note the reversal: *more* books completed corresponds to *less* time elapsed, so the inequality runs the other way. Here

$$\mathbb{E}[S_{25}] = 10(25) = 250 \text{ minutes}, \quad \text{Var}(S_{25}) = 9(25) = 225 \text{ min}^2, \quad \text{SD}(S_{25}) = 15,$$

and standardizing 240 gives

$$z = \frac{240 - 250}{15} = -\frac{10}{15} = -\frac{2}{3} \approx -0.667.$$

The event is “at most,” so it is the lower tail:

$$\begin{aligned} \mathbb{P}(\text{at least 25 books by minute 240}) &= \mathbb{P}(S_{25} \leq 240) \\ &\approx \mathbb{P}\left(Z \leq -\frac{2}{3}\right) = \Phi\left(-\frac{2}{3}\right) = \boxed{0.2525}. \end{aligned}$$